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Single-use plastics: Do we need to ban them?

The moment is right to discuss plastic use in Canada. Touching all corners of our lives, plastic products, most of which are used only once and thrown away, have been extremely prevalent in recent decades. The majority of this plastic waste ends up in the ocean and threatens the lives of marine creatures, costing billions of dollars in losses to governments and organizations.

In light of these issues, many regions and associations, including the Canadian government, are taking steps to mitigate plastic waste through bans on plastic products. Although bans are of significant importance as an effective method to solve plastic waste problems, there exist concerns about the possible negative effects of such a policy, including issues of well-being and affordability. This article examines evidence about plastic waste, especially single-use plastics, and the effects of policies trying to remediate it, to contribute to the Canadian debate surrounding Justin Trudeau's single-use plastic bans.

Issue

Plastic, as a lightweight, cheap, hygienic and low-cost material, is ever-present in our daily lives. However, its feature of being resistant and strong makes it hard to degrade, causing environmental issues, ocean pollution in particular, which poses a threat to marine creatures and humans alike. Many countries have taken – or announced to take – measures to deal with the pollution caused by plastics. Prime Minister of Canada Justin Trudeau announced his government's intention to ban single-use plastics as early as 2021¹. As it happens with most environmental issues, it is a controversial decision that has raised conflicting voices in favour and opposition for this ban. Although there are plausible alternatives for bans, I suggest here that it is necessary to implement the policy of banning single-use plastics.

The Knowledge Base

- ***The Truth about plastics.*** Plastic is a material made of synthetic or semi-synthetic organic compounds that are malleable and so can be made into solid objects. Single-use plastics, so-called disposable plastics, are plastics intended to be used only once before being thrown away or recycled. Unlike other materials, plastics are normally not biodegradable, but instead photodegradable, a process estimated to take 10-1000 years².

- ***How much plastic are we producing and consuming?*** The global production of plastics has grown dramatically. In 2014 we produced 311MT, twenty times more than in 1950. As for consumption, we observe plastic waste generation increasing annually. The tendency is similar to production (increased from 50MT in 1970 to 300MT in 2015, figure 1³). One fact worth noting is that plastic packaging (one of the single-use plastics) waste accounted for 47% of global plastic waste in 2015.
- ***How do we dispose of plastic waste?*** There are three main methods to deal with plastic waste: Recycling, incinerating and landfilling. Of these landfilling is dominant, accounting for 79% of the end of plastic waste⁴.
- ***What have other countries done?*** Many countries have taken measures to reduce the use of single-use plastics. Africa stands out as the strongest participant against plastic bags, followed by Europe⁵. Measures include regulatory instruments, economic instruments, social pressure, public-private partnership and voluntary agreements.
- ***What has Canada done?*** Leaf Rapids, Wood Buffalo, Thompson and Montreal started to ban plastics in the last decade, but there is still no information available about the impact of the measures.

About Trudeau's Policy

Justin Trudeau announced that the Government of Canada will take the following steps to reduce Canada's plastic waste:

- Ban harmful single-use plastics as early as 2021 were supported by scientific evidence and warranted;
- Work with provinces and territories to introduce standards and targets for companies manufacturing plastic products or selling items with plastic packaging so they become responsible for their plastic waste. These measures will be grounded in scientific evidence and will align, where appropriate, with similar actions being taken in the European Union and other countries.

The single-use plastics ban has been welcomed by environmental advocates, but opposition voices exist from groups worried about the implications of such ban. People with limited mobility point out that a ban on plastic straws would leave them without a better alternative; Communities living with long-term drinking water advisories⁶ are concerned about how a ban on plastic water bottles will affect their access to safe drinking water. Indeed, studies⁷ find that the primary concern of most consumers is the rising costs due to more expensive biodegradable packaging materials. Only 37.7% of surveyed people are willing to pay more for the new packaging, though 71% support a single-use plastics ban.

There are alternatives to the ban. Environmental engineer Morton Barlaz of North Carolina State University posits that it might be better to charge for single-use plastics. Environmental scientist Max Liboiron proposed to reduce oil subsidies since oil is the source of petroleum-based plastics. Both methods lead to increasing the price of single-use plastics, inducing manufacturers to look for better alternatives.

Discussion

Determining whether to implement the policy is challenging due to the complexity of the consequences. However, it's hard to deny that we should do something about the pollution. Plastic waste can cause not only environmental and economic issues but also health damage to human beings. After being landfilled, plastics decompose into small molecules and infiltrate into the soil, eventually entering into the food chain. They are highly likely to appear in the ocean due to its lightness or human's direct dumping, and eaten by marine creatures, blocking their airways and stomachs. In addition, plastic waste can cause illness or flood by blocking sewers and providing breeding grounds for pests.

Given that Canada has the longest coastline in the world, we have a unique responsibility in reducing plastic pollution, protecting the wildlife and preserving the places we love. Taking action in reducing

single-use plastics, one-third of the plastic used in Canada is a good beginning.

Quick Facts about plastic pollution

- Every year, 1 million birds and over 100,000 sea mammals worldwide are injured or die when they mistake plastic for food or become entangled⁸.
- 90% of the plastic ends up in the oceans. There are currently 50 million tons of plastic in the world's oceans⁹.
- Every day, Canadians throw away 34 million plastic bags¹⁰, and every year, over 3 million tons of plastic waste is created, representing up to \$8 billion/year in lost value.
- Only 9% of plastic waste was recycled in Canada and 87% ended up in landfills¹¹ in 2016.
- The economic cost of plastic waste is estimated to be more than \$13 billion/year.

But do we necessarily need to ban single-use plastics?

The answer is affirmative. It is important to realize that such a ban does not necessarily shrink people's welfare. According to the government's announcement, by improving plastic waste management and investing in innovative solutions, Canada can reduce 1.8 million tons of carbon pollution, generate billions of dollars in revenue, and create 42,000 jobs. Although there's no detailed explanation of those numbers, it's reasonable to say that this policy would leave us a better environment, increase revenue of innovation firms and therefore create new jobs in this industry. Savings from the cost of plastic waste disposal is a considerable number, which can be used to subsidize innovation firms or improve social infrastructure. And if a good alternative can be found, this ban would not bring inconvenience to our lives. For example, we may use a glass bottle to substitute plastic bottle.

Bans are sometimes necessary because actions without regulatory instruments can be problematic. Social pressure through campaigns against single-use plastics was used in Bangladesh during the 1990s but without regulation, more than 9 million plastic bags were wasted daily in its capital in 2002. Other campaigns enlisting public engagement to keep public spaces clean have been found unsuccessful as they cannot tackle the largest sources of pollution, which comes from garbage dumped by local residents. Other measures such as small levies on retailers that were implemented in South Africa did not prompt the desired change in consumer behaviour.

Measures such as social pressure, public-private partnership and voluntary agreements, generally have limited effects without strict government regulation. Other instruments, such as price increases, are slower in generating results. They also tend to increase inequality, as the affluent can still afford the polluting goods, now more expensive, but those at the bottom of the distribution may find themselves without options. As long as the demand for polluting goods continues, the incentives to find alternatives will be less. This means that although some reduction in plastic consumption will likely be achieved, a true change in consumer behaviour is less likely to happen.

However, bans are not free of problems. Evidence shows that bans without sufficient consultation may cause dissatisfaction in residences, and result in undesired consequences, such as what happened when the government of Rwanda banned all plastic bags in 2008. No longer after that, people started smuggling plastic bags and a lucrative black market emerged.

In addition, the existing data and reports seem insufficient to support the policy strongly. One key problem is that we still do not have a perfect substitute of plastics, which is one of the main concerns by opponents to the ban. The evidence also shows that although more than 60 countries have introduced bans and levies to curb single-use plastics, only 30% reported drastic drops in single-use plastic consumption.

Furthermore, we need to know more about the impact of these measures on the countries' economy and people's welfare. There is also a need for more data on Canada's plastic production and disposal. In a word, credible analysis of the economic benefits, including the impact on people's welfare as well as a report of reliable alternatives are needed before putting the policy into practice.

Conclusion

Neither simple bans nor actions without bans are good solutions, rather a combination of these actions seems the best approach. Banning single-use plastics offers a strong assurance of reduced plastic waste, but detailed research should be done to determine which specific product is to be banned, what are the substitutes and what the economic benefits or losses are. Sufficient consultation and education will increase compliance with the policy. Specifically, allowing for a transitional period will contribute to reducing some of the negative effects of production and consumption. It would be helpful to maintain the public informed of the progress made. In the long term, we also need a well-developed plastic waste management system to manage the remaining waste better.

Reference

- [1] Bilefsky, Dan. "Canada Plans to Ban Single-Use Plastics, Joining Growing Global Movement." The New York Times. The New York Times, June 10, 2019.
<https://www.nytimes.com/2019/06/10/world/canada/single-use-plastic-ban.html>. [2] "Environmental Issues." ThoughtCo. ThoughtCo, May 22, 2019. <http://environment.about.com/>.
- [3] Data Source: Geyer, Jambeck, and Law, 2017
- [4] Geyer, Roland, Jenna R. Jambeck, and Kara Lavender Law. "Production, Use, and Fate of All Plastics Ever Made." Science Advances. American Association for the Advancement of Science, July 1, 2017.

<https://advances.sciencemag.org/content/3/7/e1700782.full>. [5]“Feeding the Plastic Planet Minute by Minute: Anthropocene the Plastic scene.” Bellona.org, January 17, 2019. <https://bellona.org/news/industrial-pollution/2019-01-feeding-the-plastic-planet-minute-by-minute-anthropocene-the-plasticscene>.

[6] Haudenoshouty. “My Family Would Have Less Plastic Waste If We Didn’t Rely on Bottled Water for Fresh Drinking Water on Reserve

<https://t.co/eZCGKYak8F>.” Twitter. Twitter, June 10, 2019.

https://twitter.com/MOHAWKEMOTIONS/status/1138215783706705922?ref_src=twsrc^tfw

[|twcamp^tweetembed|twterm^1138215783706705922&ref_url=https://www.cbc.ca/news/indigenous/single-use-plastic-bottle-water-advisories-first-nations-1.5176370](https://www.cbc.ca/news/indigenous/single-use-plastic-bottle-water-advisories-first-nations-1.5176370).

[7]“The Single-Use Plastics Dilemma: Perceptions and Possible ...” Accessed October 10, 2019.

[https://cdn.dal.ca/content/dam/dalhousie/pdf/management/News/News & Events/Single-Use-Plastics-June-6-2019-EN.pdf](https://cdn.dal.ca/content/dam/dalhousie/pdf/management/News/News%20&20Events/Single-Use-Plastics-June-6-2019-EN.pdf).

[8]“Canada to Ban Harmful Single-Use Plastics and Hold Companies Responsible for Plastic Waste.” Prime Minister of Canada. Accessed October 10, 2019. <https://pm.gc.ca/en/news/news-releases/2019/06/10/canada-ban-harmful-single-use-plastics-and-hold-companies-responsible>.

[9] Beresford, Nick, Moeko Saito Jensen, George Edgar, Maria Sargren, United Nations Development Programme, United Nations Development Programme, and Head of European Union Delegation. “Our Plastic Problem Is out of Control. Here’s How We Can Fight It.” World Economic Forum. Accessed October 10, 2019.

<https://www.weforum.org/agenda/2018/06/how-to-fight-our-plastic-problem/>.

[10] Climate Change Canada. “Government of Canada.” Canada.ca. Government of Canada, August 8, 2019.

<https://www.canada.ca/en/environment-climate-change.html>.

[11] Climate Change Canada. “Government of Canada.” Canada.ca. Government of Canada, June 5, 2019.

<https://www.canada.ca/en/environment-climate->

change/services/managing-pollution/publications/plastic-waste-report.html.

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